Mobility-Aware UAV Positioning and Secret Key Rate Optimization for HAP–UAV CV-QKD

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***Abstract*—This provisional internal-review study examines source selection and UAV positioning for a cloud-affected HAP–UAV discrete-modulated CV-QKD model. A finite catalog of 233 exact-source ensembles defines a common location-dependent numerical rate map. An offline stationary benchmark, a current-slot Greedy policy, and finite-horizon dynamic programming (DP) are compared under identical cloud realizations and speed constraints. Across 100 paired missions, expanding the candidate set from 10 to 100 raises the mean DP penalized objective by 11.41%. At blockage probability 0.8, mean zero-rate unavailability is 24.60%, 16.53%, and 4.07% for Stationary, Greedy, and DP, respectively. At the representative penalty, DP reduces travel by 88.37% relative to zero penalty while reducing mean usable rate by 0.515%. These are conditional manuscript-model outputs: the recorded two-dimensional mutual information and homodyne Holevo expression do not specify a consistent Bob measurement. The results characterize the saved numerical optimization and do not establish physical quantum security.**

***Index Terms*—Continuous-variable quantum key distribution, UAV, high-altitude platform, cloud blockage, probabilistic shaping, dynamic programming.**

Internal review only. Security-model correction and author review remain pending. Status:
`UNVALIDATED_SECURITY_MODEL`;
`NOT_READY_FOR_PUBLICATION_SCALE_RUNS`.

I. HAP–UAV CV-QKD System and Problem Formulation

A. Mobility-Aware UAV Positioning and SKR Optimization

We consider one UAV at a fixed altitude over a $1\,\mathrm{km}\times 1\,\mathrm{km}$ area, discretized into J candidate positions $\mathcal{Q}=\{\mathbf{q}_j\}_{j=1}^{J}$. The mission has N_t slots of duration Δt. Let $c_{j,n}\in\{0,1\}$ indicate cloud blockage (one means blocked). The matrix $\mathbf{C}=[c_{j,n}]$ contains spatially and temporally correlated Bernoulli states. P2 and P3 know the complete matrix and are offline benchmarks; Greedy observes only the current slot.

1) Source and Stationary-Position Optimization: For an unblocked link, the recorded manuscript-style raw rate is

$$K_j(\nu,V_A)=\beta_{\mathrm{rec}}I^{2D}_{AB,j}(\nu,V_A)-\chi_{BE,j}(\nu,V_A). \tag{1}$$

Equation (1) is retained as a conditional numerical metric, subject to the measurement-consistency limitations below. For a finite exact-source catalog $\mathcal{G}$, define

$$\mathcal{P}_1:\quad K_j^{\star}=\left[\max_{(\nu,V_A)\in\mathcal{G}}K_j(\nu,V_A)\right]^{+}, \tag{2}$$

where $[x]^{+}=\max\{x,0\}$. The raw maximum is selected before clipping. P1 is solved once per location under stationary clear-sky statistics. The source associated with the current position is available to every method. Complete blockage gives $S_{j,n}=(1-c_{j,n})K_j^{\star}$ without another quantum-channel evaluation. The stationary benchmark is

$$\mathcal{P}_2:\quad j_{\mathrm{fix}}^{\star}=\arg\max_{j}\frac{1}{N_t}\sum_{n=1}^{N_t}S_{j,n}. \tag{3}$$

Thus, the best fixed position depends on both its clear-sky rate and its full cloud sequence. Its total travel distance is zero.

2) Mobility-Aware Trajectory Optimization: Let j_n be the position in slot n and $d_{i,j}=\|\mathbf{q}_i-\mathbf{q}_j\|_2$. We solve

$$\begin{aligned}\mathcal{P}_3:\quad \max_{j_1,\ldots,j_{N_t}}\quad & \frac{1}{N_t}\sum_{n=1}^{N_t}S_{j_n,n}-\frac{\lambda}{D_{\mathrm{ref}}}\sum_{n=2}^{N_t}d_{j_{n-1},j_n}\\ \text{subject to}\quad & d_{j_{n-1},j_n}\le v_{\max}\Delta t,\quad n\ge 2.\end{aligned} \tag{4}$$

Here $D_{\mathrm{ref}}>0$ normalizes distance, and $\lambda\ge 0$ has units bit/use. Staying is permitted. The initial position is free, with no terminal constraint, deployment cost, or return travel. Distance is a mobility-cost surrogate rather than calibrated propulsion energy.

B. Recorded Physical-Layer Model and Security Boundary

Alice sends the coherent-state ensemble $\{p_k,\alpha_k\}_{k=1}^{256}$. With $[\hat{x},\hat{p}]=2i$ and vacuum quadrature variance one, the normalization is $V_A=2\sum_k p_k|\alpha_k|^2=2\bar{n}$. The source moments C_s,w_s are computed from this same ensemble; the subscript distinguishes them from the cloud matrix. For clear-sky power transmittance η, the saved evaluator uses

$$\begin{aligned}\eta_f&=\langle\sqrt{\eta}\rangle^2,\\ N=\eta_f\xi_f&=(\langle\eta\rangle-\eta_f)V_A+\xi_{\mathrm{ch}}\langle\eta\rangle,\\ V&=1+V_A,\qquad W=1+\eta_f V_A+N,\\ Z&=2\sqrt{\eta_f}C_s-\sqrt{2Nw_s}.\end{aligned}$$

The convention is $\xi_n=\xi_{\mathrm{ch}}$. The additive N is output-referred excess noise; loss-induced vacuum is already included in W's unit baseline. The active detector is ideal (efficiency one, electronic noise zero). No imperfect trusted-detector claim is made. The clear-sky distribution retains a separate angle-of-arrival outage atom at $\eta=0$ and its active density on $[0,1]$, without renormalizing. Moment reduction at the covariance level has a precedent for Gaussian modulation [1]; it is not an identity for the full discrete-input fading likelihood.

Specifically, the saved MI kernel uses the normalized likelihood

$$p_{\mathrm{num}}(\mathbf{y}|k) = \frac{\exp[-\|\mathbf{y} - 2\sqrt{\eta_f}(\Re\alpha_k, \Im\alpha_k)\|^2/(2(1+N))]}{2\pi(1+N)}$$

and integrates both real components to obtain I_{AB}^{2D}. The Holevo branch uses the standard-form covariance with entries V, W, Z and the *homodyne* conditional eigenvalue

$$\nu_3^{\mathrm{hom}} = \sqrt{V(V - Z^2/W)}.$$

These two branches do not specify one consistent single-mode Bob measurement: a homodyne outcome has one quadrature, while joint measurement of both quadratures adds measurement vacuum. The measurement-dependent Holevo expressions are explicitly different in [2]. Consequently, all reported "usable SKR" values are conditional outputs of the recorded manuscript model, not established achievable secret-key rates.

Further, evaluating only Z_- does not establish the worst-case Holevo value over all admissible correlations; the general interval-maximization caveat in [2, Sec. 4, footnotes 11–12] also applies. A consistent measurement/noise specification, admissible correlation treatment, parameter estimation, and fading/outage conditioning protocol require author review. No correction is applied to the historical numerical results in this manuscript revision.

II. Optimization Method

A. Finite-Catalog Search and Stationary Benchmark

The selected source parameters satisfy

$$(\nu_j^\star, V_{A,j}^\star) \in \arg\max_{(\nu,V_A)\in\mathcal{G}} K_j(\nu, V_A). \tag{5}$$

Every catalog candidate is evaluated at each distinct channel state. Exact source moments are reused only for their authenticated ensembles. Source-independent channel moments can be shared, while N and ξ_f are recomputed for each V_A. Ties use increasing ν, V_A, and ensemble hash after decreasing raw rate. The result is optimal within $\mathcal{G}$, with no continuous-source claim. P2 compares all J full-horizon means in $O(JN_t)$ time; position ties select the smallest index.

B. Greedy Trajectory Baseline

The current-slot policy starts at

$$j_1^G \in \arg\max_j S_{j,1}. \tag{6}$$

For $n \geq 2$, it selects

$$j_n^G \in \arg\max_{j: d_{j_{n-1}^G, j} \leq v_{\max}\Delta t} \left\{ S_{j,n} - \frac{N_t\lambda}{D_{\mathrm{ref}}} d_{j_{n-1}^G, j} \right\}. \tag{7}$$

The factor N_t makes this score consistent with (4). The decision function receives only the current rate vector and previous position, together with the transition graph and fixed parameters. It cannot read future clouds. Its worst-case time is $O(N_tJ)$, and ties use the smallest index. Greedy and DP use identical rate maps, clouds, speed limits, and distance penalties.

Algorithm 1 Source Search and Offline UAV Positioning

Input: $\mathcal{Q}, \mathcal{G}, \mathbf{C}, N_t, \Delta t, v_{\max}, D_{\mathrm{ref}}, \lambda$

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for each position j do
    Select the maximum raw catalog rate in (5).
    Set K_j^⋆ by (2) and S_{j,n} = (1 − c_{j,n}) K_j^⋆.
end for
Select P2 using (3).
Compute Greedy with (6)–(7).
Initialize F_1 using (9).
for n = 2, ..., N_t do
    for each position j do
        Update F_n(j) and save B_n(j) by (10)–(11).
    end for
end for
Select the best final position and backtrack.
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Output: P1 sources, P2 position, Greedy path, DP path, and metrics.

C. Dynamic Programming for UAV Trajectory Optimization

Define the feasible predecessors

$$\mathcal{N}(j) = \{i \in \{1, \ldots, J\} : d_{i,j} \leq v_{\max}\Delta t\}. \tag{8}$$

Let $F_n(j)$ be the maximum cumulative reward for trajectories ending at j in slot n. Multiplying (4) by N_t gives

$$F_1(j) = S_{j,1}, \qquad j = 1, \ldots, J, \tag{9}$$

and, for $n \geq 2$,

$$F_n(j) = S_{j,n} + \max_{i\in\mathcal{N}(j)} \left\{ F_{n-1}(i) - \frac{N_t\lambda}{D_{\mathrm{ref}}} d_{i,j} \right\}. \tag{10}$$

Store a maximizing predecessor

$$B_n(j) \in \arg\max_{i\in\mathcal{N}(j)} \left\{ F_{n-1}(i) - \frac{N_t\lambda}{D_{\mathrm{ref}}} d_{i,j} \right\}. \tag{11}$$

The optimal objective is

$$U^\star = \frac{1}{N_t} \max_j F_{N_t}(j). \tag{12}$$

Starting at a maximizing final position, backtracking uses $j_{n-1}^\star = B_n(j_n^\star)$. The implementation stores F_n/N_t, so its reward is $S_{j,n}/N_t$ and its edge cost is $\lambda d_{i,j}/D_{\mathrm{ref}}$; this is exactly the same recursion. DP is globally optimal for the given finite grid and rate matrix. Its time is $O(N_tJ^2)$, backpointers use $O(N_tJ)$ memory, and the dense distance/transition matrices additionally use $O(J^2)$. Since fixed and Greedy paths are feasible, $U_{\mathrm{DP}} \geq U_{\mathrm{P2}}, U_{\mathrm{Greedy}}$. This statement concerns the penalized objective, not mission-wise SKR or availability.

III. Numerical Results and Discussion

A. Simulation Setup

The synthetic HAP altitude is 20 km and its horizontal position is $(11419.6551, 450)$ m. The UAV flies at 1 km over 100 cell centers, with coordinates $50, 150, \ldots, 950$ m along each axis (the x index changes first). The reference mission has 30 slots of 20 s and $v_{\max} = 15$ m/s, permitting transitions

TABLE I
RETAINED SYNTHETIC SIMULATION PARAMETERS

Parameter	Value
Wavelength; full beam divergence	1550 nm; 233 μrad
Receiver diameter; zenith transmission	0.08 m; 0.92
Angular jitter; horizontal jitter	15.61 μrad; 1 m
Boresight displacement	0 m
AoA standard deviation; FOV half angle	1.5 mrad; 6.25 mrad
Hufnagel–Valley wind speed	21 m/s
Ground-level $C_n^2(0)$	$10^{-14}\,\mathrm{m}^{-2/3}$
β_{rec}; ξ_{ch}	0.95; 0.0186 SNU
Detector efficiency; electronic noise	1; 0
Default p; ℓ; latent ρ	0.4; 250 m; 0.9
Independent mission IDs	$0, \ldots, 99$
Reference distance D_{ref}	1000 m

up to 300 m. These are simulation assumptions rather than measured deployment data.

The common P1 catalog contains 233 authenticated sources: 197 Maxwell–Boltzmann (MB) and 36 uniform ensembles. P1 selects $(\nu, V_A) = (0.125, 8)$ at all 100 positions, where $V_A = 8$ is the upper MB-catalog boundary. Thus, differences among positioning methods do not arise from changing the selected shaping parameters. The saved clear-sky rates range from 1.16296×10^{-4} to 1.57800×10^{-4} bit/use. Table I lists the retained optical and noise parameters. No numerical model or source calculation is rerun for this revision.

The saved complete-blockage model thresholds a stationary Gaussian field: its spatial covariance is $\Sigma_{ij} = \exp(-d_{i,j}/\ell)$ and $\mathbf{z}_n = \rho \mathbf{z}_{n-1} + \sqrt{1-\rho^2}\boldsymbol{\epsilon}_n$, where $\boldsymbol{\epsilon}_n \sim \mathcal{N}(0, \Sigma)$ and $\mathbf{z}_1 \sim \mathcal{N}(0, \Sigma)$. Blockage is $c_{j,n} = \mathbb{I}\{z_{j,n} < \Phi^{-1}(p)\}$. The seed is 20261015 with mission IDs 0–99. Configurations share innovations for paired comparisons. Latent ρ and ℓ are distinct from binary-cloud correlations. Unless varied, $(p, \ell, \rho) = (0.4, 250\,\mathrm{m}, 0.9)$.

For a path $\mathbf{j} = (j_1, \ldots, j_{N_t})$, we report

$$\overline{K}(\mathbf{j}) = \frac{1}{N_t} \sum_{n=1}^{N_t} S_{j_n,n}, \tag{13}$$

$$D(\mathbf{j}) = \sum_{n=2}^{N_t} d_{j_{n-1},j_n}, \tag{14}$$

$$U(\mathbf{j}) = \overline{K}(\mathbf{j}) - \lambda D(\mathbf{j})/D_{\mathrm{ref}}. \tag{15}$$

Availability is $N_t^{-1} \sum_n \mathbb{I}\{S_{j_n,n} > 0\}$; unavailability is its complement, expressed in percent. This nonzero-rate definition has no positive engineering threshold. The reference penalty is the fixed absolute value $\lambda_{\mathrm{ref}} = 9.468026677248516 \times 10^{-6}$ bit/use. The ten saved penalties have multipliers $\{0, 10^{-4}, 3\times 10^{-4}, 10^{-3}, 3\times 10^{-3}, 10^{-2}, 3\times 10^{-2}, 0.1, 0.3, 1\}$ of $\lambda_{\mathrm{hi}} = 0.003156008892416172$ bit/use; thus $\lambda_{\mathrm{ref}} = 0.003\lambda_{\mathrm{hi}}$.

Means use 100 independent missions per configuration. Pointwise 95% percentile-bootstrap intervals resample whole missions with 20000 replicates; paired method contrasts use the same mission IDs. Saved intervals are retained, and missing tradeoff intervals use saved metrics and existing resampling counts. These intervals quantify mission variability, not quantum-security uncertainty. The multi-curve presentation is informed by [3]; its satellite model and numerical results are not transferred to this experiment.

B. Impact of Candidate UAV Positions

Figure 1 compares deterministic nested subsets $J = 10, 20, 40, 60, 80, 100$ at λ_{ref}. The mean DP objective increases from 1.39004×10^{-4} to 1.54863×10^{-4} bit/use, an 11.41% increase relative to $J = 10$. P2 increases from 1.36348×10^{-4} to 1.50243×10^{-4}, and Greedy from 1.07800×10^{-4} to 1.48311×10^{-4} bit/use. Adding positions preserves existing feasible paths, so optimal P2 and DP objectives cannot decrease; Greedy has no such general guarantee.

At $J = 20$, DP reaches 96.71% of its $J = 100$ objective. However, the $J = 10$ transition graph has eight connected components and seven isolated positions, whereas the sampled subsets are connected from $J = 20$ onward. The initial improvement combines expanded spatial choices and connectivity changes; it does not identify an economically optimal grid size or isolate candidate density alone.

C. Impact of Cloud Blockage Probability

Figure 2(a) shows decreasing mean usable SKR as p increases from 0.1 to 0.8 for all three methods. DP decreases from 1.57663×10^{-4} to 1.43475×10^{-4} bit/use. The probability masks are nested under common innovations: increasing p removes usable rewards. Therefore, the optimal P2 and DP *objectives* cannot increase; a general availability or Greedy ordering does not follow.

The DP-minus-P2 mean objective difference grows from 1.11215×10^{-7} to 2.72806×10^{-5} bit/use. Dividing each difference by its corresponding P2 mean gives gains of 0.071% and 25.042%. At $p = 0.8$, the paired 95% interval for the absolute difference is $[2.46760, 2.99896] \times 10^{-5}$ bit/use. The larger relative gain is consequently accompanied by a larger absolute gain. No interior maximum occurs on the evaluated probability grid, and no claim is made beyond $p = 0.8$.

In Fig. 2(b), unavailability at $p = 0.8$ is 24.60% for P2, 16.53% for Greedy, and 4.07% for DP. These observed availability benefits do not mean that P3 maximizes availability. Greedy's mean objective is below P2 at $p = 0.1$–0.4; its paired differences at $p = 0.5$ and 0.7 have intervals containing zero. At $p = 0.8$, its objective exceeds P2 by 8.54160×10^{-6} bit/use, less than DP's improvement.

D. Impact of Temporal Cloud Correlation

Figure 3 varies latent $\rho \in \{0, 0.3, 0.6, 0.9, 0.98\}$ at $p = 0.4$, $\ell = 250$ m, and λ_{ref}. From $\rho = 0$ to 0.98, DP's mean objective increases from 1.44559×10^{-4} to 1.56806×10^{-4} bit/use, while its mean distance decreases from 1038.25 to 31.84 m. Its relative objective gain over P2 falls from 23.712% to 0.155%.

Saved-mask diagnostics support a persistence explanation. The cloud-free fraction at the selected P2 location rises from 77.03% to 99.97%, and the mean number of locations clear throughout the 30-slot mission rises from zero to 29.56. The global fraction of adjacent cell-slot states that change falls

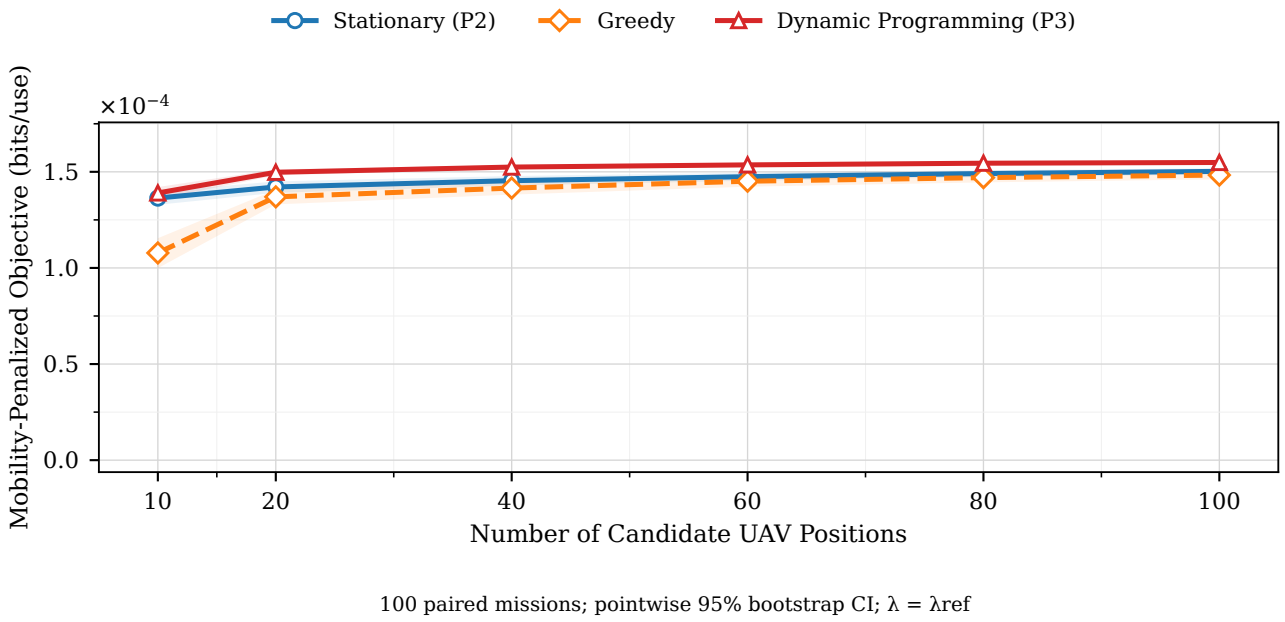


Fig. 1. Mean mobility-penalized objective versus the number of candidate UAV positions at λ_{ref}. Stationary (P2), Greedy, and DP use deterministic nested subsets of the original coordinates and common cloud missions. Shaded bands show pointwise 95% whole-mission bootstrap intervals. The objective includes a distance penalty and is distinct from average usable SKR.

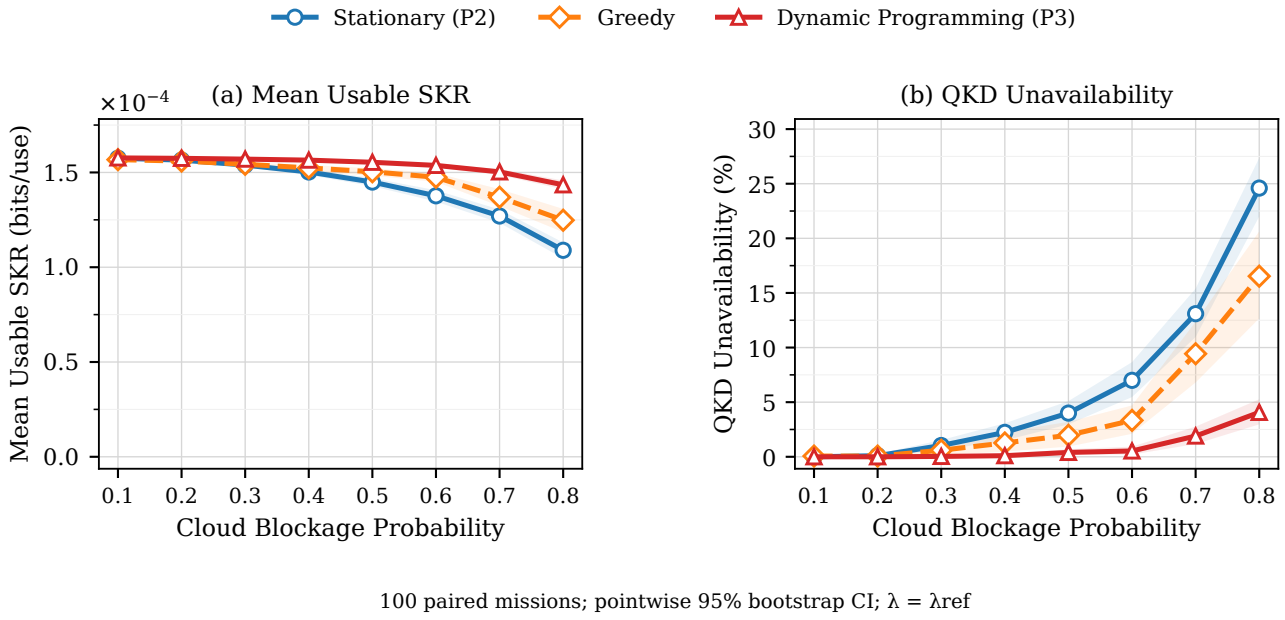


Fig. 2. Cloud-blockage sensitivity at λ_{ref}: (a) mean usable SKR and (b) zero-SKR unavailability for P2, Greedy, and DP. All eight probabilities $p = 0.1, 0.2, \ldots, 0.8$ use 100 paired missions with common innovations. Bands are pointwise 95% whole-mission bootstrap intervals. Unavailability counts zero usable SKR, without an engineering threshold; neither panel is the optimized penalized objective.

from 48.21% to 6.35%. DP's mean number of moves decreases from 7.36 to 0.19; missions containing DP movement decrease from 100 to 10 out of 100.

Persistent favorable locations therefore make offline stationary selection more competitive in this experiment. This is a supported mechanism, not an isolated causal result or universal monotonic law. P2 and DP choose initial positions using future clouds, whereas Greedy does not. At $\rho = 0.98$, the latent correlation time $-20/\log\rho \simeq 990$ s exceeds the 600-s mission. Empirical blockage varies across the finite samples despite fixed nominal p.

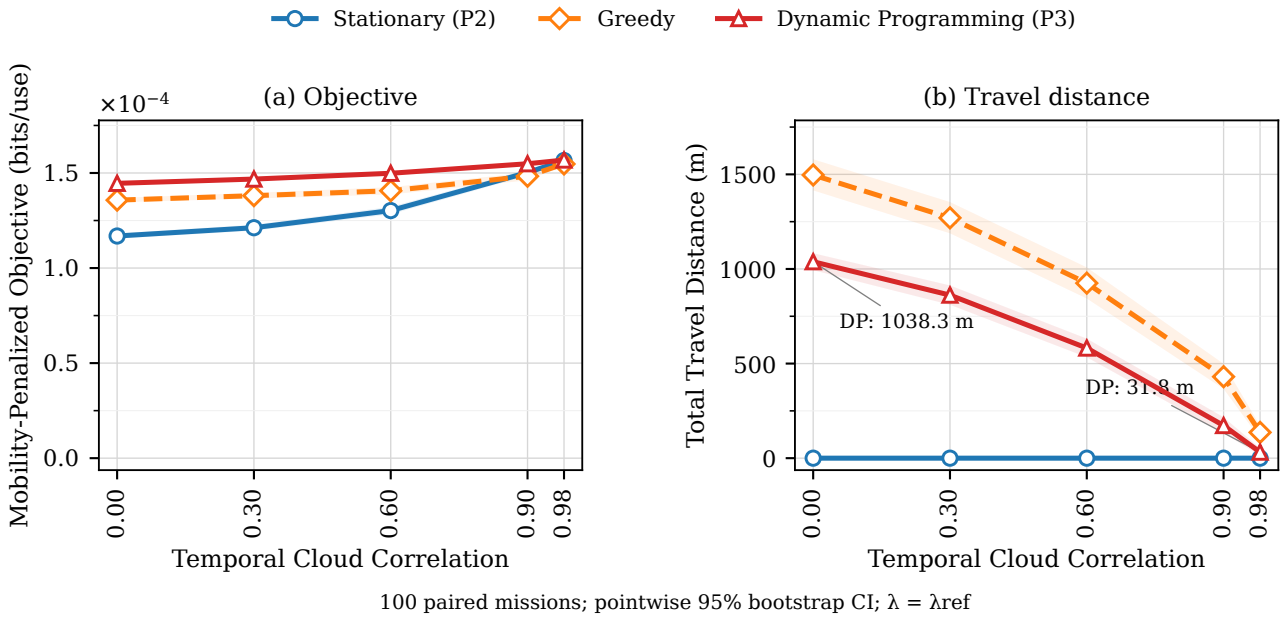


Fig. 3. Latent temporal-correlation sensitivity: (a) mean penalized objective and (b) mean total travel distance at $p = 0.4$, $\ell = 250$ m, and λ_{ref}. P2, Greedy, and DP share 100 mission IDs and common innovations. P2 and DP know the full horizon; Greedy uses the current slot. The latent coefficient differs from binary blockage correlation. Bands are pointwise 95% whole-mission bootstrap intervals.

E. SKR–Mobility Tradeoff

Figure 4 connects saved policies in the original order of ten increasing penalties. At $\lambda = 0$, DP achieves mean usable SKR 1.57305×10^{-4} bit/use and distance 1482.69 m. At λ_{ref}, distance falls to 172.43 m and mean SKR to 1.56495×10^{-4} bit/use. Relative to the respective zero-penalty means on the same 100 default missions, these changes are an 88.37% distance reduction and a 0.515% SKR reduction.

At λ_{ref}, Greedy travels 430.61 m, while DP has both higher mean SKR and higher objective. DP's SKR gains relative to P2 and Greedy are 4.162% and 2.695%; its objective gains are 3.075% and 4.418%, respectively. The paired DP-minus-P2 objective interval is $[3.56503, 5.73139] \times 10^{-6}$ bit/use. The distinction matters because movement is penalized only in U.

At sampled penalties $0.1\lambda_{\mathrm{hi}}$ and above, all saved DP paths are stationary and coincide with P2 in objective. This does not identify a continuous stopping threshold. Greedy can stop at a worse location because it chooses its initial position from the first slot. Optimal DP distance is nonincreasing with λ for the same feasible set and rewards; the plotted points are evaluated numerical operating choices, not global physical Pareto frontiers.

F. Discussion and Limitations

Across the investigated parameter ranges, the largest mean-SKR span occurs for candidate count in DP, speed in Greedy, and blockage probability in P2. These are descriptive ranges over unequal domains, not normalized sensitivity indices. Among cloud sweeps, the largest relative DP-versus-P2 objective gain occurs at $p = 0.8$, while the largest absolute difference occurs at $\rho = 0$. The supplementary spatial sweep also distinguishes absolute performance from mobility gain: increasing ℓ from 50 to 1000 m reduces absolute performance but increases the DP-minus-P2 objective difference from 2.25288×10^{-6} to 8.10831×10^{-6} bit/use. Full-grid blockage grows from zero to 2.70% of slots. A larger relative gain therefore does not imply increasing spatial diversity.

P1 provides the common catalog-selected map, P2 captures offline fixed-position selection, and DP-minus-P2 measures the extra value of permitting movement in that setting. Since all positions select the same source, cloud-avoidance gains cannot be assigned to adaptive shaping. These three methods do not isolate matched-energy shaping gain or stationary-selection gain relative to a fixed deployment. DP-minus-Greedy includes differences in future-state and initial-position information. Objective dominance does not imply higher SKR in every positive-penalty mission.

The study is restricted to a finite source catalog and UAV grid, synthetic weather and geometry, and complete cloud blockage. Tracking interruptions, reacquisition, deployment, return travel, and calibrated propulsion energy are omitted. The mixed 2D-MI/homodyne model in Section I-B, the effective-likelihood approximation, and correlation-domain treatment prevent a validated physical SKR interpretation. Trusted receiver noise would require a separate consistent model [4]. Asymptotic rate arithmetic and mission confidence intervals establish neither operational nor finite-key, composable, or unconditional security.

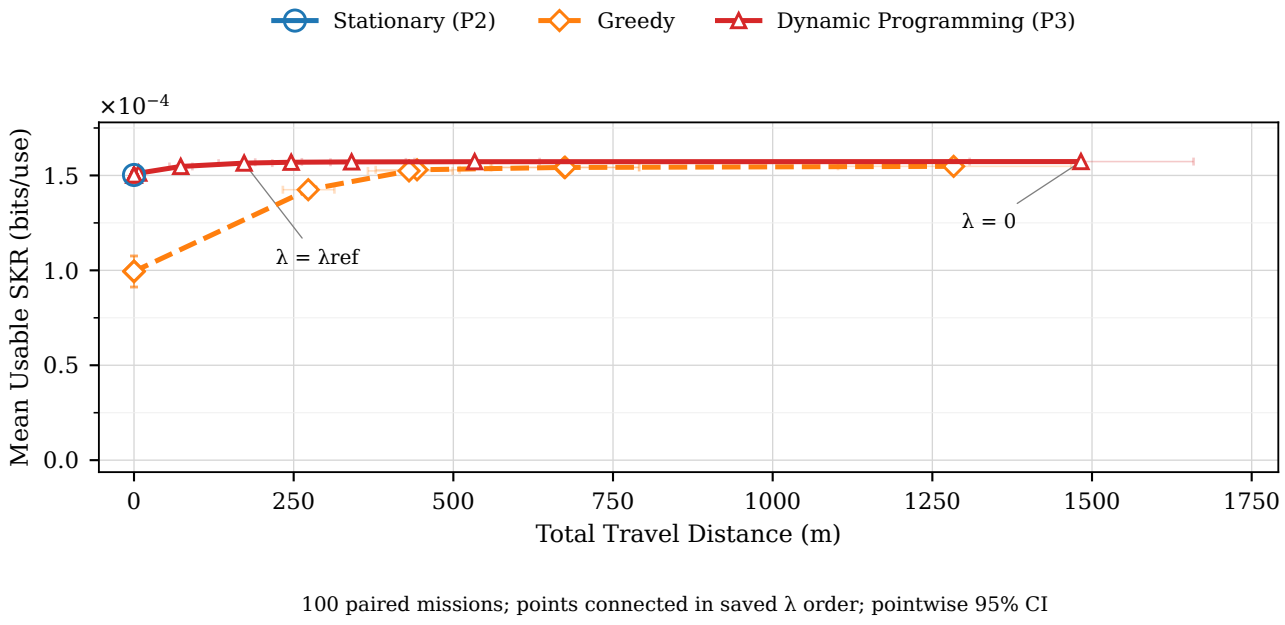


Fig. 4. Evaluated usable-SKR–distance tradeoff for Greedy and DP, with P2 as the stationary reference, under default clouds. Curves connect the ten saved penalties in their original increasing order; DP annotations identify zero and representative penalties. Error bars are pointwise 95% intervals for each mean, not joint confidence regions. Coincident high-penalty DP points and the P2 reference are retained. These conditional model outputs are not claimed to be global physical Pareto frontiers.

IV. Conclusion

Finite-catalog source selection and finite-grid positioning provide a reproducible optimization framework for the recorded HAP–UAV numerical model. Offline DP maximizes the stated SKR–distance objective for known clouds, while P2 selects a fixed location and Greedy uses current-slot information. Increasing candidate count improves the optimal objective, strong blockage increases the observed benefit of mobility, and persistent favorable cloud states reduce its incremental value over offline stationary selection. At the representative penalty, DP gains 3.075% in mean objective over P2 and 4.418% over Greedy on the default missions; relative to zero-penalty DP it reduces travel by 88.37% with a 0.515% decrease in mean usable rate. These conclusions apply to the finite catalog, synthetic clouds, and distance surrogate. A measurement-consistent MI/Holevo model, justified correlation-domain bound, and operational fading/outage security analysis are prerequisites for interpreting these conditional numerical rates as secure key generation. This manuscript remains provisional for internal review.

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